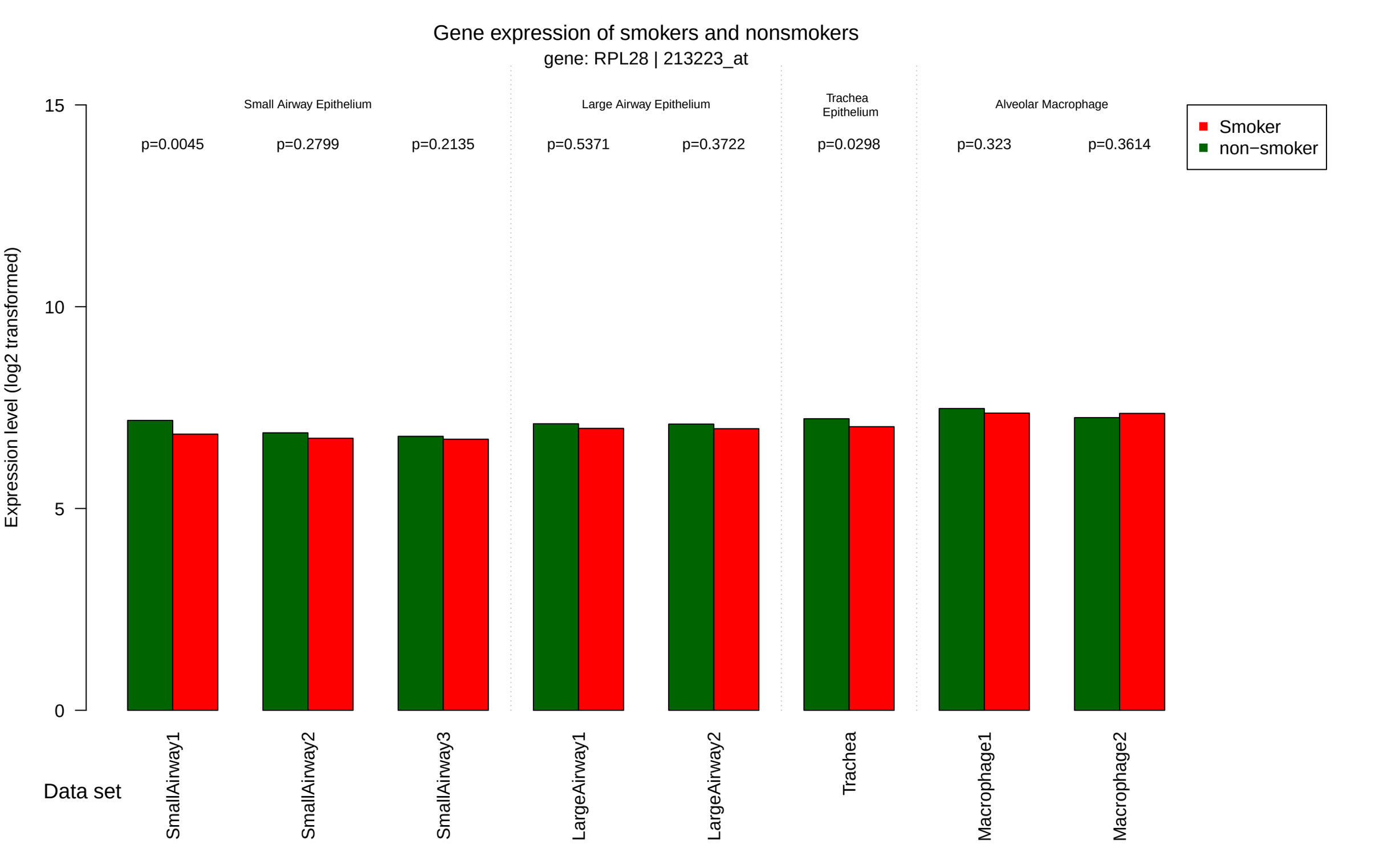
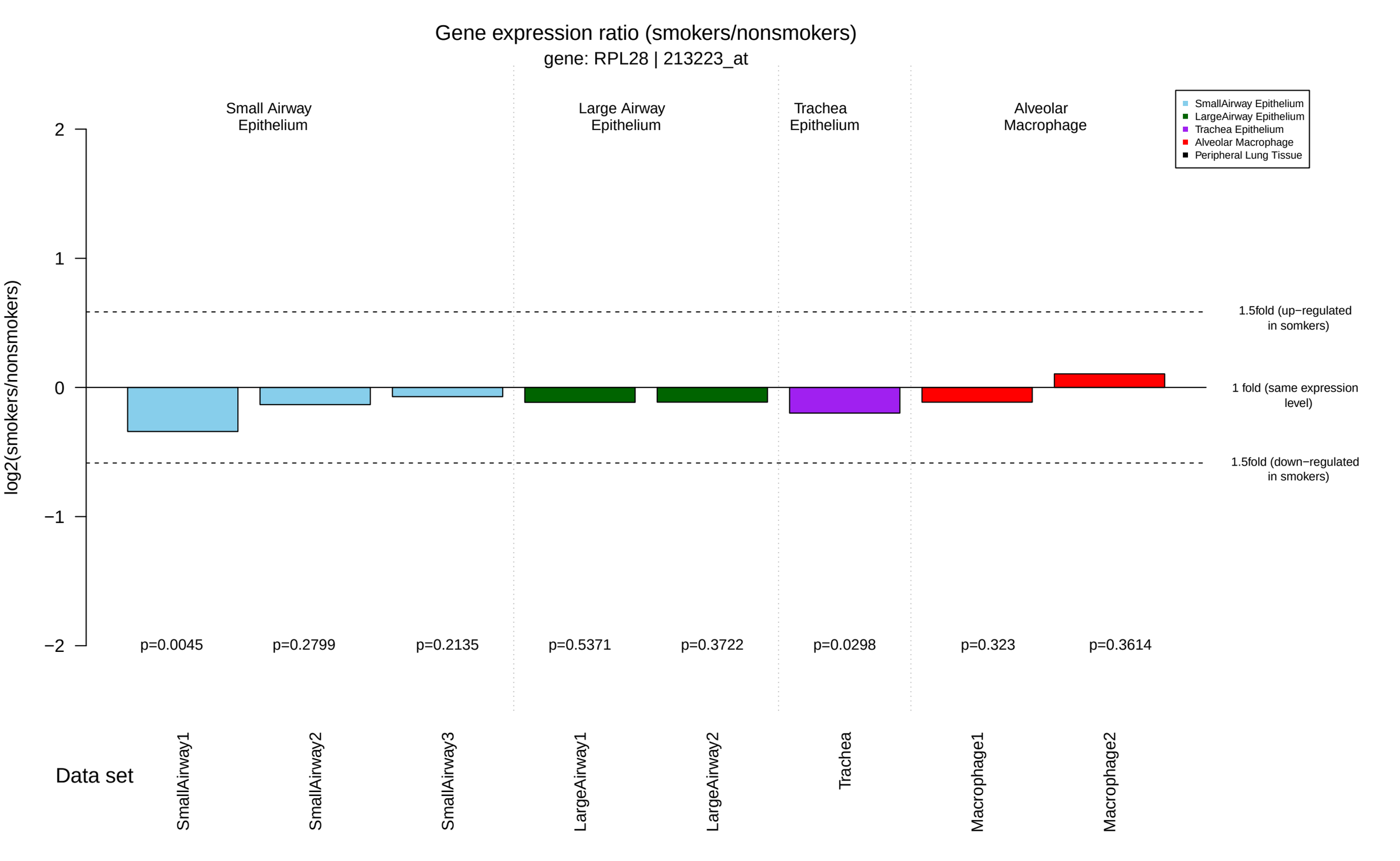
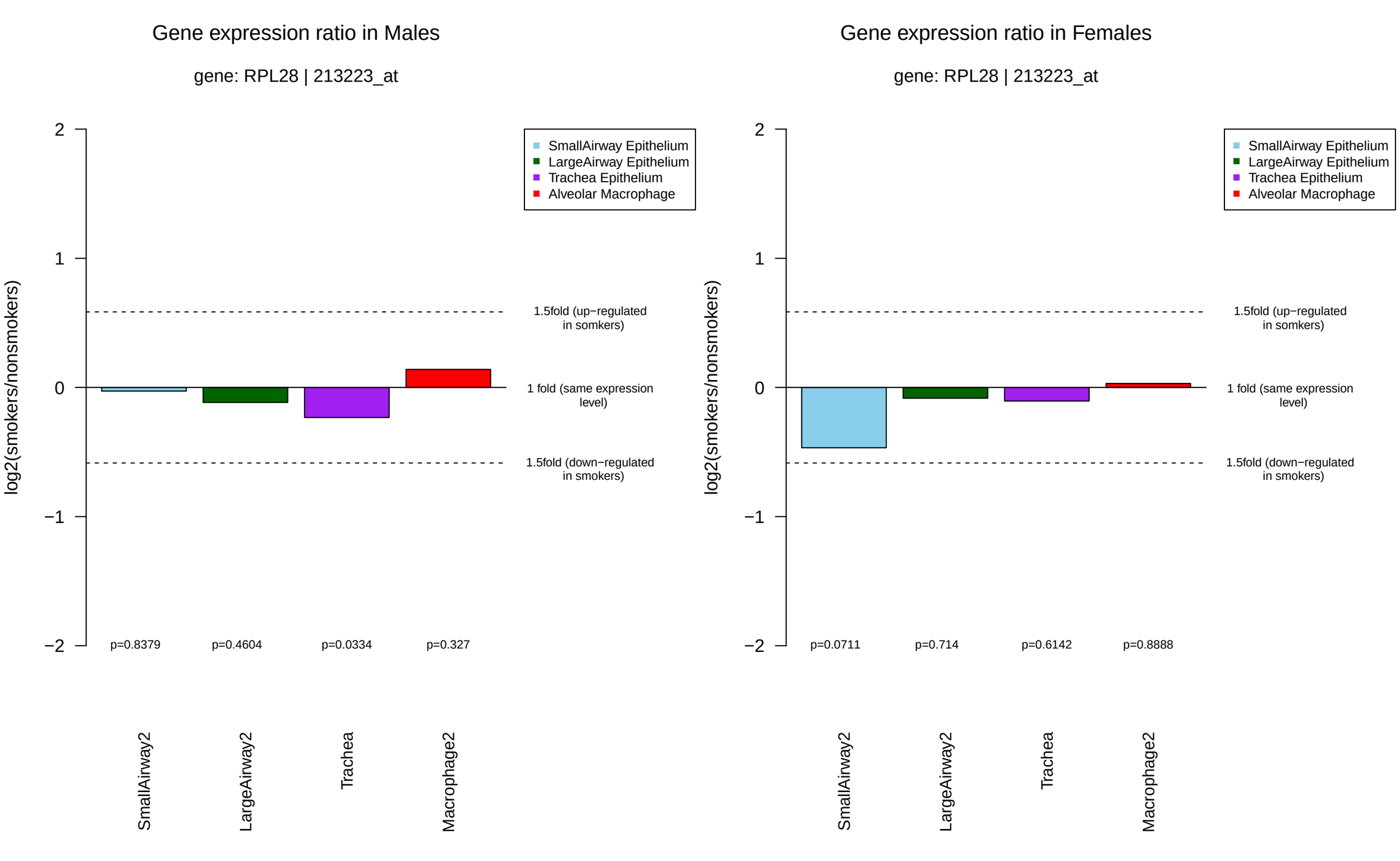
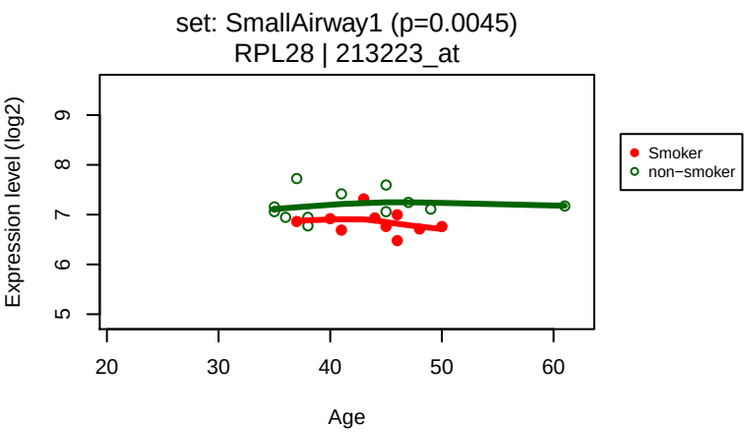
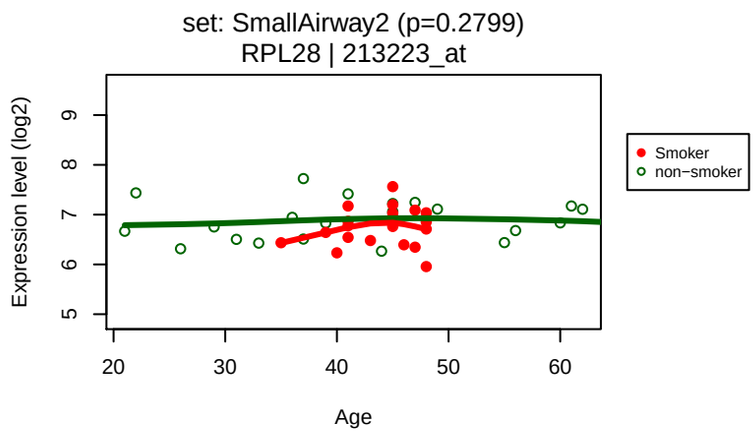
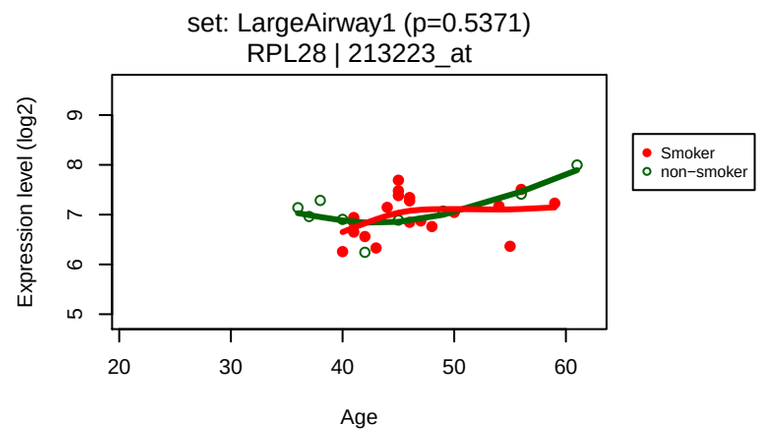

Gene expression of smokers and nonsmokers
gene: RPL28 | 213223_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0045
p=0.2799
p=0.2135
p=0.5371
p=0.3722
p=0.0298
p=0.323
p=0.3614
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

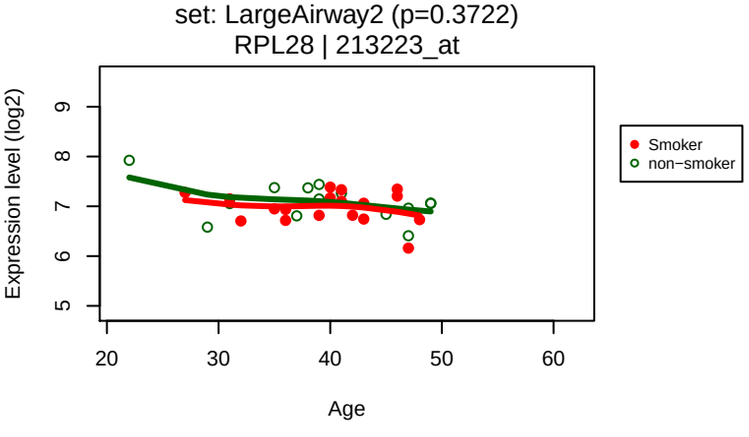

Gene expression ratio (smokers/nonsmokers)
gene: RPL28 | 213223_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0045
p=0.2799
p=0.2135
p=0.5371
p=0.3722
p=0.0298
p=0.323
p=0.3614
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

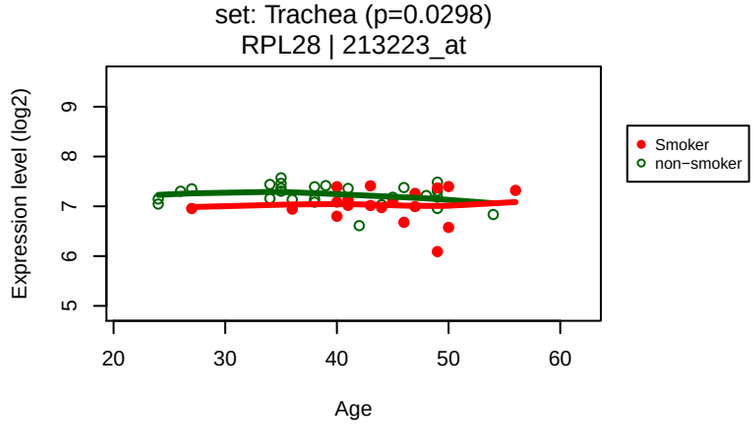

Gene expression ratio in Males
gene: RPL28 | 213223_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8379
p=0.4604
p=0.0334
p=0.327
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPL28 | 213223_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0711
p=0.714
p=0.6142
p=0.8888
SmallAirway2
LargeAirway2
Trachea
Macrophage2

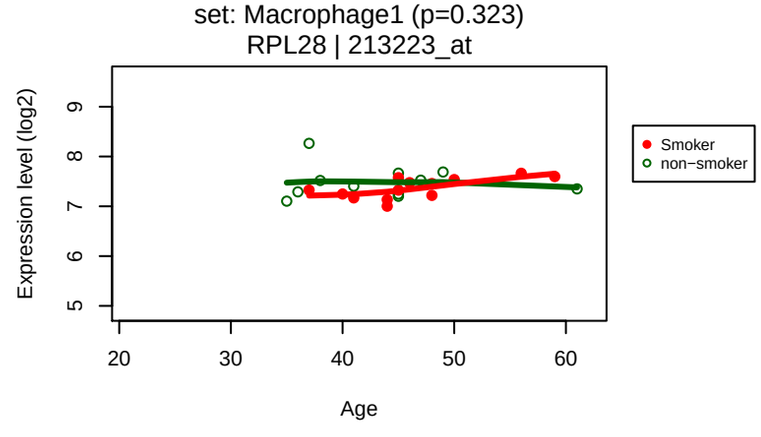

set: SmallAirway1 (p=0.0045)
RPL28 | 213223_at
Expression level (log2)
Age
Smoker
non−smoker

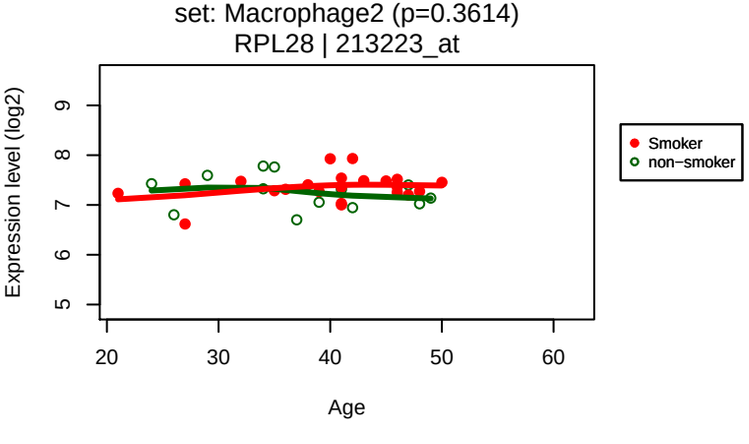

set: SmallAirway2 (p=0.2799)
RPL28 | 213223_at
Expression level (log2)
Age
Smoker
non−smoker

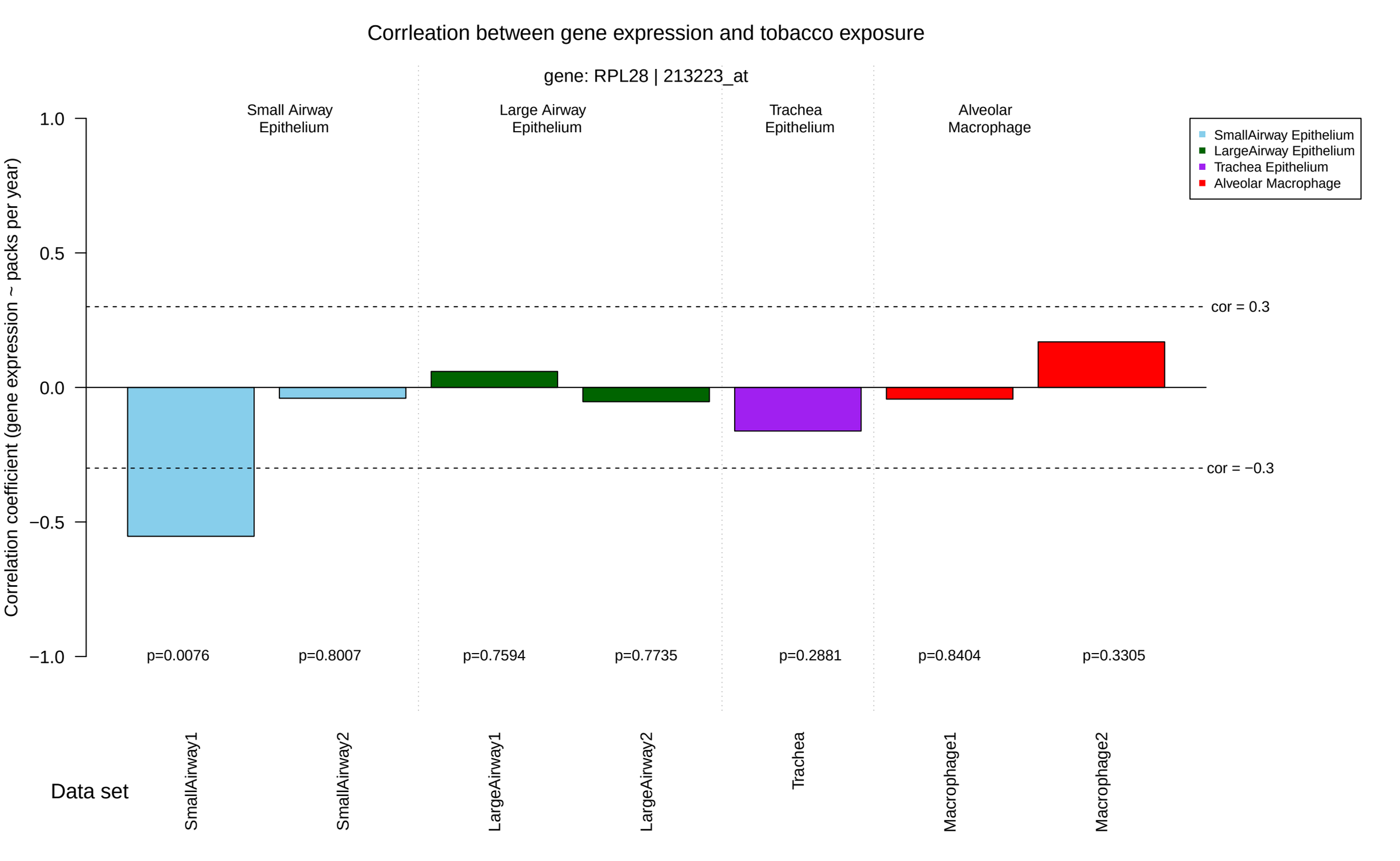

set: LargeAirway1 (p=0.5371)
RPL28 | 213223_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3722)
RPL28 | 213223_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0298)
RPL28 | 213223_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.323)
RPL28 | 213223_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3614)
RPL28 | 213223_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPL28 | 213223_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0076
p=0.8007
p=0.7594
p=0.7735
p=0.2881
p=0.8404
p=0.3305
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2